

HOW MUCH DOES RECOVERY FINE-TUNING RECOVER? OPERATING-POINT-DEPENDENT GAINS IN PRUNED TEXT-TO-AUDIO DIFFUSION

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ABSTRACT

Structured pruning of text-to-audio diffusion models is followed by recovery fine-tuning, whose benefit is usually reported as one score at one inference setting. Using the released pruned and fine-tuned AudioLDM-M checkpoints at two pruning severities (65% and 83% of U-Net parameters removed), we measure the recovery gain in text–audio alignment with a paired, common-noise protocol across clip duration and prompt domain. The gain is large at the fine-tuning duration of 10.24s (CLAP 0.055 $\rightarrow$ 0.299 at 83% pruning), several-fold smaller at 3.84s (0.015 $\rightarrow$ 0.100), and absent on held-out music. The duration dependence is pre-registered, replicated on a disjoint prompt set, robust to the pruning-seam convention, and corroborated by a second scorer, event-level metrics and a frame-level grounding model, which shows the native-duration gain spread across the whole clip. Recovery should be reported across operating points; lacking a matched dense fine-tuned control and human ratings, our claims concern evaluation, not mechanism.

Index Terms— text-to-audio generation, diffusion models, structured pruning, recovery fine-tuning, evaluation

1. INTRODUCTION

Structured pruning followed by recovery fine-tuning is the standard route to cheaper diffusion models [1–3], recently applied to text-to-audio (TTA) generation with AudioLDM [4, 5]. The recovery stage is judged by the restoration of one aggregate number—a Fréchet distance or a KL divergence on the in-domain test set at one sampler setting. A generative model, however, is used at an inference operating point: a clip duration, a prompt domain, a sampler configuration. Nothing guarantees that a recovery gain measured at one point describes the fine-tuned checkpoint at another.

We ask: does the gain of recovery fine-tuning over the pruned checkpoint depend on the operating point at which it is evaluated? We take the released pruned and fine-tuned AudioLDM-M-Full checkpoints of Singh et al. [5] at two pruning severities as the object of study, train nothing, and measure the paired recovery gain in text–audio alignment across two clip durations and two prompt domains. Our findings are:

- **Duration.** The recovery gain is several times larger at the fine-tuning duration (10.24s) than at 3.84s. The interaction is pre-registered and replicated on a disjoint prompt set at the stronger severity, seam-robust, and holds on a scale-free (win-rate) statistic at both severities.

- **Domain.** At matched duration the gain is present in-domain (AudioCaps) and absent on held-out music at both severities and at both durations.
- **Where the gain lives.** A frame-level grounding model shows the native-duration gain spread uniformly over the clip, and a crop analysis shows the short-duration deficit arises from generating short clips, not from scoring short excerpts: the first 3.84s of a native generation already carries the whole gain at severity 1 and over half of it at severity 2.

2. BACKGROUND

Pruning and recovery in diffusion models. Structural pruning of diffusion backbones with fine-tuning recovery is established for image models [1–3]. For TTA, Singh et al. [5] L1-prune the AudioLDM U-Net, fine-tune on AudioCaps for 10^6 steps and report in-domain FAD/KL at a fixed sampler; notably their fine-tuned pruned model surpasses the unpruned model in-domain (FAD 1.57 vs. 3.95), a first sign that the recovery stage also acts as further in-domain specialisation. All these works evaluate recovery at one operating point.

Evaluating text-to-audio generation. CLAP similarity [6] is the standard automatic proxy for text–audio alignment; distribution metrics such as FAD [7] are sensitive to sample size and embedding choice [8, 9]; event classifiers [10], human-aligned CLAP [11] and frame-level grounding models [12] measure other axes, motivating a paired, multi-metric design with one pre-specified primary endpoint.

Duration as an operating point. Latent audio diffusion conditions generation on the latent length and is trained at a fixed duration (10.24s for AudioLDM [4, 13]; newer systems condition on timing explicitly [14]), so duration is a mechanistically motivated factor for a checkpoint fine-tuned at one duration; fine-tuning also trades in-distribution gains for out-of-distribution losses [15], motivating the domain axis.

3. EXPERIMENTAL DESIGN

3.1. Systems

The dense model is AudioLDM-M-Full [4] (U-Net stage multipliers (1, 2, 3, 5), 415.96 M U-Net parameters), itself AudioCaps-fine-tuned by its authors. We study the two released pruning severities of [5]: severity 1, (1, 2, 3, 1), 145.67 M parameters (−65.0%), and severity 2, (1, 2, 1, 1), 71.08 M (−82.9%; the authors report 39% fewer multiply–accumulates at this budget). At each severity we compare

the pruned checkpoint (P) with the fine-tuned checkpoint (P+FT), the released result of 10^6 steps of AudioCaps recovery fine-tuning of P [5] (EMA weights materialised, checksums verified). Since the released pruned checkpoints carry no usable EMA weights, P is the released L1 channel selection applied to the dense EMA weights: at severity 1 this reproduces the released raw pruned weights bit-exactly (690/690 tensors); at severity 2 the released checkpoint differs in three decoder-seam tensors, so both conventions are evaluated (A' primary, B' sensitivity). "Recovery" names the fine-tuning stage, not an achieved restoration; we train no system.

3.2. Operating points and prompt batteries

Two durations: the native point (10.24s, latent length 256; the fine-tuning duration) and a short point (3.84s, latent length 96). Two domains: in-domain AudioCaps test captions [16] and held-out hip-hop music captions from MusicCaps [17], a sub-domain with near-zero exposure in the recovery corpus. Severity 2 uses 192 AudioCaps prompts (disjoint from all severity-1 and training prompts) and 64 music prompts (3 replicates at 3.84s, plus 10.24s at severity 2); severity 1 uses 96 AudioCaps prompts for the pre-registered domain test, a matched 80-prompt subset for the duration comparison, and 64 music prompts at 3.84s. Generation uses DDIM [18], 50 steps, guidance 2.5, $\eta = 0$, fp32, single generation, EMA weights (off the published recipe, Sec. 5).

3.3. Paired protocol and scoring

Within each operating point the compared systems receive identical generation noise x_T per prompt (common random numbers), so a per-prompt contrast is not confounded by sampling noise; across durations the latent shapes differ, so duration contrasts are prompt-paired but not noise-paired. The primary scorer is fused CLAP (laion/clap-htsat-fused, rev. 365dea6e) [6], which repeat-pads 3.84s audio to 10s and centre-crops 10.24s audio to 10s; every (system, operating point) is scored with one fixed-order, fixed-seed call (part of the endpoint). The unit is the prompt (music replicates averaged first); intervals are 95% prompt-level percentile bootstrap intervals ($B = 10^4$).

3.4. Quantities and analysis plan

Let R be the mean per-prompt recovery gain, CLAP(P+FT) - CLAP(P), at a given (domain, duration); R_{nat} , R_{short} and R_{mus} denote AudioCaps-native, AudioCaps-short and music-short. The duration interaction is

$$J = R_{\text{nat}} - R_{\text{short}}, \quad (1)$$

a paired per-prompt contrast with pre-specified gate $\log_5(J) > 0$; the domain contrast at matched duration is $R_{\text{short}} - R_{\text{mus}}$; the smallest effect size of interest is 0.025 (analysis plan and registration status: Table 1).

4. RESULTS

4.1. The recovery gain grows with clip duration

Fig. 1 and Table 2 give the primary result. At severity 2, on the disjoint 192-prompt set with estimands frozen before scoring, the interaction is clearly resolved, $J =$

Table 1. Analysis plan ("Reg.": estimand, gate and prompts committed before any score was seen).

Analysis	Reg.	Role
Sev. 1 domain test (3.84s)	yes	pre-registered; failed
Sev. 1 duration J	yes	follow-up; gate $\log_5 > 0$
Sev. 2 J , R_{nat} , R_{short} , R_{mus} , seam B'	yes	primary; disjoint set
Sev. 2 music at 10.24s	yes	design completion; no gate
Human-CLAP, KL, PANNs, FAD	yes	secondary; no gate
FineLAP frame-level profile	yes	post-result diagnostic
Win-rate J ; crop analysis	no	post-hoc sensitivity

+0.159 [+0.131, +0.187]: fine-tuning raises CLAP alignment from 0.055 to 0.299 at the native duration ($R_{\text{nat}} = +0.244$) but only from 0.015 to 0.100 at 3.84s ($R_{\text{short}} = +0.085$). As duration slopes, the pruned checkpoint gains just +0.040 [+0.029, +0.052] from the longer clip, the fine-tuned one +0.200 [+0.173, +0.226]. The result is seam-robust (B': $J = +0.161$). The gain is smaller at the short point, not absent (R_{short} resolved; 90% CI [+0.069, +0.102] outside ± 0.025).

At the milder severity 1 the same pattern appears with smaller effects: $R_{\text{short}} = +0.008 [-0.023, +0.039]$, $R_{\text{nat}} = +0.052 [+0.009, +0.093]$, $J = +0.044 [-0.001, +0.087]$, whose interval narrowly includes zero on the raw cosine scale. Because every system scores higher at 10.24s (dense: 0.204 $\rightarrow$ 0.352), a raw-scale difference-in-differences could reflect scale alone, so as a post-hoc sensitivity check we also report the interaction in the fraction W of prompts on which P+FT beats P (Table 2). This scale-free interaction is resolved at both severities: 0.44 $\rightarrow$ 0.64 at severity 1 ($\Delta W = +0.20 [+0.06, +0.34]$), 0.72 $\rightarrow$ 0.87 at severity 2 (+0.15 [+0.07, +0.22]); the second scorer agrees at severity 1 ($J_{\text{HC}} = +0.075 [+0.012, +0.137]$). Severity-2 effects are larger on every axis, but the severities come from different experiments and prompt sets, so that comparison is descriptive.

4.2. At matched duration the gain is domain-specific

Held-out music separates the domains at fixed duration. At severity 2 and 3.84s the gain is resolved in-domain ($R_{\text{short}} = +0.085$) and null on music ($R_{\text{mus}} = +0.009 [-0.013, +0.032]$; $W = 0.48$), a matched-duration domain contrast of +0.076 [+0.047, +0.105]. At severity 1 the pre-registered domain test (96 prompts) found no in-domain gain at 3.84s ($R_{\text{AC}} = -0.002 [-0.027, +0.021]$) and a negative music contrast ($R_{\text{mus}} = -0.094 [-0.124, -0.065]$), a domain contrast of +0.092 [+0.054, +0.131]. The music penalty is severity-specific; the absence of a music gain replicates. At the native duration the music contrast is likewise null (+0.005 [-0.028, +0.039]; $W = 0.53$), so the large native gain is confined to the fine-tuning domain: the 10.24s domain contrast is +0.239 [+0.195, +0.283].

4.3. Where in the clip the gain lives

Two post-result analyses locate the native-duration gain. First, as a prospectively frozen diagnostic, we score the native clips with FineLAP [12], a frame-level language-audio grounding model (EAT encoder, not CLAP-derived) that reports per 0.16s frame how strongly the requested event is grounded. Fine-tuning raises grounding across the whole

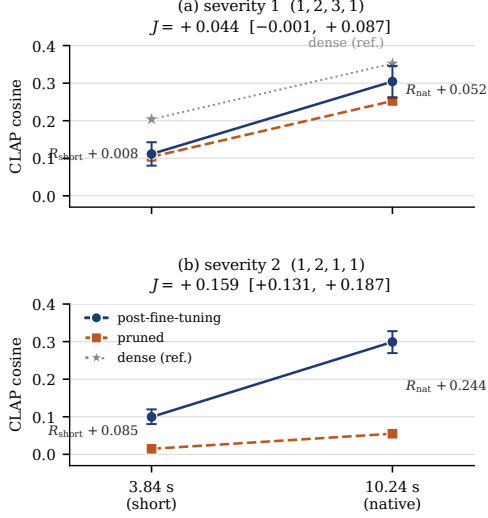


Fig. 1. The recovery gain grows with clip duration. Mean CLAP cosine of the pruned (P, dashed) and fine-tuned (P+FT, solid) checkpoints at the short and native points for (a) severity 1 and (b) severity 2; whiskers: 95% CI of the paired gain R , drawn about the P+FT mean; J annotated. The dense model (grey, panel a; unpaired reference) also scores far higher at the native duration: the short point is off-native for the whole family.

Table 2. Recovery gain in CLAP alignment: mean cosine per checkpoint, paired gain $R = \text{CLAP}(P+FT) - \text{CLAP}(P)$ with 95% CI, and fraction W of prompts on which P+FT beats P. †: CI excludes 0; ‡: pre-registered gate passes. J paired per prompt; domain contrasts two-sample. n : sev.1 AudioCaps 80 (96 for the domain contrast), sev.2 AudioCaps 192, music 64.

Setting	P	P+FT	R [95% CI]	W
Severity 1, (1, 2, 3, 1)				
AudioCaps 3.84s	0.104	0.111	+0.008 [-0.023, +0.039]	0.44
AudioCaps 10.24s	0.253	0.304	+0.052† [+0.009, +0.093]	0.64
music 3.84s	0.117	0.023	-0.094‡ [-0.124, -0.065]	0.20
$J = R_{\text{nat}} - R_{\text{short}}$			+0.044 [-0.001, +0.087]	
domain 3.84s ($n=96$)			+0.092‡ [+0.054, +0.131]	
Severity 2, (1, 2, 1, 1)				
AudioCaps 3.84s	0.015	0.100	+0.085† [+0.066, +0.105]	0.72
AudioCaps 10.24s	0.055	0.299	+0.244† [+0.215, +0.273]	0.87
music 3.84s	0.005	0.014	+0.009 [-0.013, +0.032]	0.48
music 10.24s	0.089	0.094	+0.005 [-0.028, +0.039]	0.53
$J = R_{\text{nat}} - R_{\text{short}}$			+0.159‡ [+0.131, +0.187]	
domain 3.84s			+0.076† [+0.047, +0.105]	
domain 10.24s			+0.239† [+0.195, +0.283]	

clip (severity 2: semantic mass +0.27 [+0.22, +0.33], occupancy +0.27, quarter coverage +0.38, peak +0.41; CIs exclude 0; both severities; seam-robust), and the gain in the first 3.84s equals the gain in the remainder ($D_{\text{early}} = +0.275$ vs. $D_{\text{late}} = +0.273$; $T = -0.002 [-0.024, +0.020]$, Fig. 2a): the native gain is not confined to the extra seconds. Second, a post-hoc crop analysis separates generation length from scoring window: we score only the first 3.84s of each 10.24s generation with the same CLAP convention and com-

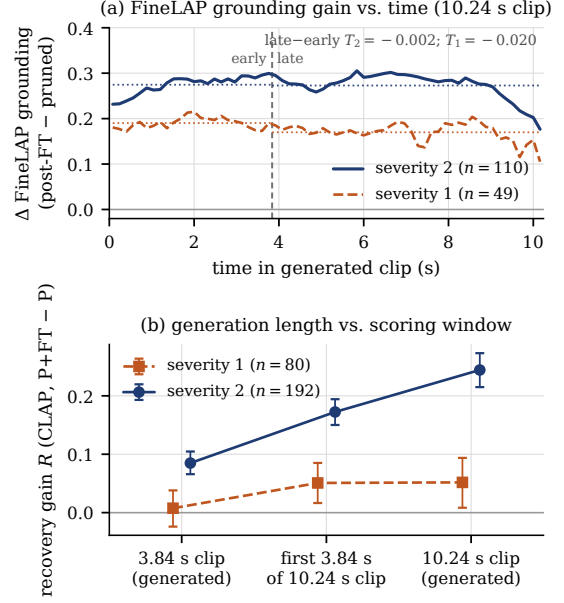


Fig. 2. (a) FineLAP frame-level grounding gain (P+FT - P) vs. time in the 10.24s clip (dotted: window means; vertical line: 3.84s): uniform, not back-loaded. (b) Recovery gain R (95% CI) on the generated 3.84s clip, on the first 3.84s of the 10.24s clip, and on the full 10.24s clip: the crop carries the whole gain at severity 1 and over half at severity 2—a generation-length, not a scoring-window, effect.

pare with the separately generated 3.84s clips of the same prompts (Fig. 2b). The crop carries a large part of the native gain. At severity 1, $R_{\text{crop}} = +0.051 [+0.016, +0.085]$ equals R_{nat} ($R_{\text{nat}} - R_{\text{crop}} = +0.001 [-0.027, +0.029]$) and exceeds R_{short} by +0.043 [+0.002, +0.085]; at severity 2, $R_{\text{crop}} = +0.172 [+0.150, +0.194]$, with $R_{\text{crop}} - R_{\text{short}} = +0.087 [+0.065, +0.110]$ and $R_{\text{nat}} - R_{\text{crop}} = +0.072 [+0.055, +0.089]$ (seam-robust). The short-duration deficit is thus mostly a property of what the fine-tuned checkpoint generates when asked for a short clip, not of scoring a short excerpt—consistent with a checkpoint specialised to its fine-tuning duration.

4.4. Corroboration and pre-registered negatives

CLAP is the sole primary endpoint; secondaries corroborate without gating. At the severity-2 native point Human-CLAP [11], a CLAP model fine-tuned on human ratings (automatic, not a listening test), gives $R_{\text{nat}} = +0.375 [+0.340, +0.408]$ and $J = +0.185$; two pre-specified event-level metrics favour P+FT with paired CIs excluding zero—KL to real references 2.23 vs. 4.45 ($\Delta = +2.22 [+1.93, +2.53]$) and PANNs top-10 captured labels 1.46 vs. 0.60 ($\Delta = +0.86 [+0.70, +1.02]$); FAD agrees (6.92 vs. 27.4, descriptive).

Three pre-registered hypotheses failed and are kept. (i) The hypothesis that motivated the study—that recovery fine-tuning trades in-domain for out-of-domain alignment—requires an in-domain gain at 3.84s and found none ($R_{\text{AC}} \approx 0$, Sec. 4.2), which reframed the work from mechanism to

evaluation. (ii) The severity-1 music penalty did not replicate (Human-CLAP alone shows a small negative music contrast at severity 2, -0.037 $[-0.068, -0.005]$; the sign is unestablished). (iii) A “late allocation” account of the duration effect is rejected by FineLAP ($T \approx 0$). Finally, P+FT is not restored to dense: at severity 1 and 10.24s the dense model leads P by $+0.099$ $[+0.058, +0.140]$ and P+FT by $+0.048$ $[-0.000, +0.096]$ —neither a resolved gap nor equivalence.

5. DISCUSSION AND LIMITATIONS

Across severities, scorers and metrics, the recovery gain of the released fine-tuned AudioLDM-M checkpoints tracks the operating point of the fine-tuning itself: it is largest at the fine-tuning duration, confined to the fine-tuning domain, and heterogeneous across prompts (Table 2, column *W*). This is consistent with the recovery stage acting, at least in part, as further in-domain, native-duration specialisation—the reading also suggested by the fine-tuned pruned model surpassing the unpruned one in-domain in [5]. Post-pruning recovery is thus better described as multi-operating-point behaviour than as a single restored score. We recommend that recovery studies report (1) the training-matched operating point and one motivated off-native point (another duration or domain); (2) absolute scores of both P and P+FT at each point; (3) paired gains with prompt-level uncertainty under common random numbers; and (4) an explicit interaction across points—an implication of one case study (one model family, two severities, two durations, two domains).

Limitations. Mechanistic attribution is blocked: the matched control—the dense model given the same AudioCaps fine-tune—was trained by the original authors but no longer exists, and the public dense text-fine-tuned release is not equivalent (different data and recipe), so the dependence cannot be attributed to pruning rather than to fine-tuning in general. Primary inference rests on CLAP-family scorers (FineLAP is the only non-CLAP model) and there is no human evaluation (a designed listening study was cancelled before launch). Sampler settings differ from the published recipe (DDIM 50 vs. 200, guidance 2.5 vs. 3.5, single vs. best-of-3), so external validity to it is untested; both systems are always evaluated identically. Only two durations were tested; the frame-level and crop analyses are post-result diagnostics.

6. CONCLUSION

For the released pruned and fine-tuned AudioLDM-M checkpoints, recovery fine-tuning recovers much at its own operating point and much less away from it: the CLAP gain is several times larger at the fine-tuning duration than at 3.84s, absent on held-out music, and uniform across the clip. A single in-domain, native-duration score overstates what the fine-tuned checkpoint delivers elsewhere. Audio examples are online.¹

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¹<https://gbibbo.github.io/audioldm-modality-swap-pruning/>